

Transfer of Micro-LEDs with Roll-Based Direct Overlay Alignment for Manufacturing Transparent Displays

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Transparent displays are crucial for various applications, particularly for their potential use as windows in future self-driving cars. These displays require high transparency, low power consumption, and high mechanical reliability. Micro-LEDs have emerged as ideal devices for the transparent displays. Efficient mass-production processes are essential for the commercialization of transparent micro-LED displays. This study presents roll-based mass transfer to enhance the productivity of transparent micro-LED displays. Roll transfer processes traditionally face resolution challenges in alignment repeatability and positional errors in both the transverse direction (TD) and machine direction (MD). This study proposes a roll-to-plate (R2P) transfer process with overlay alignment to improve the repeatability precision of the alignment. Detailed experimental analyses address positional errors in the TD and MD, attributed to initial contact errors and linear velocity asynchrony, respectively. The results demonstrate successful micro-LED transfer onto a transparent circuit board (TCB) with a maximum positional error of 3.2 μm and a 99.75% yield. The resulting micro-LED display achieves a transparency of 72.5% with 68 pixels per inch. This study overcomes the alignment challenges in the R2P process and contributes to the commercialization of transparent micro-LED displays. It is expected to positively impact the manufacturing of transparent applications that involve rolling processes.

This highlights the need for high-resolution displays with enhanced visibility, transparency, and flexibility.^[1–3] Micro-LEDs are promising pixel technologies for these displays and exhibit distinct advantages over conventional LCDs and OLEDs. These advantages include wide viewing angles, low power consumption, high resolution, and robust mechanical reliability. Consequently, these features render micro-LEDs particularly suitable for applications that require outdoor or transparent display functionalities.^[4–6] To use micro-LEDs as pixels in a display, a transfer process is required to transfer the micro-LEDs to a printed circuit board (PCB). In addition, the transfer process of micro-LEDs should be performed with high productivity for mass-production.^[7,8]

Several micro-LED transfer methods have been proposed, including the electrostatic method, laser-assisted method, elastomeric flat stamp method, fluidic self-assembly method, and the micro-vacuum transfer method.^[1,9,10] The electrostatic method employs a sophisticated

electrostatic chuck for selective pickup and placement of micro-LEDs,^[11,12] but it has limitations in the transfer area and involves expensive chucks.^[1,4] The laser-assisted transfer method utilizes a dynamic release layer for high-speed transfer, but it suffers from positional drift and low transfer force.^[13–17] The elastomeric flat stamp method relies on rate-dependent adhesion for the picking and placing process, yet it also faces challenges related to limited transfer area.^[18–26] The fluidic self-assembly leverages fluidic momentum and small-scale forces to assemble micro-LEDs in liquid, but it experiences low transfer yield and positional uncertainty.^[27–30] Lastly, the micro-vacuum transfer method utilizes pressure differences to pick and place micro-LEDs, but it suffers from limitations in transfer area and expensive micro-vacuum chucks.^[10]

Despite these advancements in the transfer methods, challenges remain in commercializing micro-LED displays including positional uncertainty, costly chuck/stamp expenses, and limited transfer area. To address these issues, we have investigated a roll-to-plate (R2P) transfer method. R2P manufacturing processes offer high-speed and large-area capabilities, making them suitable for various applications, such as microchip transfer, reverse offset printing, and nanoimprinting.^[31–36] For R2P transfer,

1. Introduction

In the evolving landscape of applications, such as autonomous vehicles and virtual reality/augmented reality (VR/AR) technologies, a diverse range of display technologies is imperative.

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micro-LEDs can be transferred by optimizing the pressure on the roll without relying on the speed dependence of the roll stamp,^[37] which requires a load control system that can evenly apply the load. This allows for high-speed transfer at a rate of 36 million per hour^[3] without the transfer yield being dependent on the roll speed. Because micro-LEDs are transferred through the R2P process, the efficiency improves as the number of micro-LEDs to be transferred increases. Moreover, the small contact area between the roll stamp and plate is advantageous for pressure control, and there is nearly no limitation on the transfer area. However, the R2P process has limitations in selectively transferring micro-LEDs, and ensuring accuracy and precise repeatability is challenging.^[38] In roll-based processes, positional errors include machine direction (MD) errors along the roll rotation direction, and transverse direction (TD) errors perpendicular to the roll rotation direction.^[39–44] While the above studies address positional errors in the R2P reverse offset printing process and the methods to resolve them, the R2P transfer of microdevices involves the same mechanism, leading to the occurrence of similar positional errors. However, in R2P micro-LED transfer processes, micro-LEDs have unique dimensions in the thickness direction, which distinguish them from conventional printing processes. Special considerations include precise alignment, high-resolution printing technology, and precision transfer systems; however, there is little research on the causes and solutions of positioning errors. Another alignment issue is repeatability. This problem is because of the nature of the roll equipment, which requires measuring the position of the device attached to the roll stamp and the substrate placed on the plate, and then virtually aligning the positions through calculation. Repeatability, in turn, depends on the precision of the roll equipment hardware or the accuracy of the alignment calculations.

In this study, we introduce a roll-based direct overlay alignment (RDOA) method to address the repeatability precision. This method is applicable only when the final substrate is transparent. This requires placing a transparent circuit board (TCB) on a transparent quartz plate and aligning it by simultaneously observing the TCB and micro-LEDs attached to the roll stamp using a bottom camera. In addition, we identified the causes of the MD and TD position errors through experiments and provided a simple solution. This provides a new strategy and solution for the alignment problem in future micro-LED transfers, contributing to their applicability to large area and/or transparent display applications.

2. Results and Discussion

2.1. Process of R2P Transfer

To fabricate transparent micro-LED displays, a transfer process is necessary to move the micro-LED arrays on the interposer to the TCB. In this study, micro-LEDs were transferred using the R2P technology. A schematic of the micro-LED R2P transfer process is shown in **Figure 1a**. Initially, an interposer with a micro-LED array was placed on the plate. As shown in **Figure S1a** (Supporting Information), the micro-LED array on the interposer is 160×120 , and the size of each micro-LED is $32 \mu\text{m} \times 40 \mu\text{m}$, with the LEDs in a face-down state. Face-down implies that the electrodes of the micro-LED are directed toward the interposer side. After aligning

the micro-LEDs on the interposer, force was applied to pick up the micro-LED array using a roll stamp. The micro-LEDs picked up by the roll stamp were in a face-up state, indicating exposure of the micro-LED electrodes, as shown in **Figure S1b** (Supporting Information). After completing the micro-LED array pickup using the roll stamp, the TCB was positioned on a quartz plate, aligning its electrodes with the electrode positions of the micro-LEDs. Subsequently, the roll stamp contacted the TCB, a load was applied, and the transfer process was initiated. Through the R2P process, a transparent display was demonstrated by transferring the micro-LEDs, as shown in **Figure 1b** and **Movie S1** (Supporting Information). There is a difference between the pixel resolution at which the micro-LEDs are transferred, and the resolution observed in images and movies, because it drives clusters of 32×30 . Meanwhile, in the manufacturing of the TCB, only the red micro-LED could be activated, owing to the current leakage occurring between the upper and lower electrodes, as well as among the red, green, and blue electrodes. As shown in **Figure 1c**, the transparency is 72.5% in the visible spectrum of 450–700 nm.

The transfer yield was measured by evaluating 25 regions of the TCB to which the micro-LEDs were transferred, as shown in **Figure S2a** (Supporting Information). **Figure S2b** (Supporting Information) shows images of the 25 regions of the interposer, roll stamp, and TCB. The number of LEDs in this image was counted to calculate the picking and transfer yields, as presented in **Table S1** (Supporting Information). The picking yield was measured 100%, and the transfer yield is $\approx 99.75\%$. We also needed to calculate the lighting yield. However, due to the current TCB configuration where the 5×4 array operates as a single cluster, accurately determining the lighting yield was challenging. This is because if one micro-LED in a cluster has a problem, it could affect others, making precise calculation of yield impossible. The positional error of the micro-LEDs was measured by evaluating three regions of the TCB, as shown in **Figure S3a** (Supporting Information). On the interposer, MD average errors ranged from -0.70 to $0.78 \mu\text{m}$, and TD average errors ranged from $0.32 \mu\text{m}$ to $0.80 \mu\text{m}$. On the TCB, MD average errors varied from -2.80 to $2.57 \mu\text{m}$, and TD average errors ranged from $-3.10 \mu\text{m}$ to $2.17 \mu\text{m}$ across different areas.

In the case of the R2P transfer process, the average yield of transfer typically reaches 99.99%.^[1,3] However, in this study, it was slightly lower at 99.75%, which can be attributed to the presence of particles or dust in the transfer process as their size exceeded that of the micro-LEDs. While the experiment was conducted in a 10000-class cleanroom, the transfer faced challenges owing to particles or dust. Additionally, when conducting R2P micro-LED transfer using the same method, it can be observed that a transfer yield of over 99% is achieved as shown in **Figures S4 and S5** (Supporting Information). Furthermore, the maximum error of $3.2 \mu\text{m}$ is linked to the micro-LED's placement on the interposer, which has an inherent error of $\approx 3.2 \mu\text{m}$. Nevertheless, this $3.2 \mu\text{m}$ error accounts for only 10% of the micro-LED size and this is acceptable for the TCB designed in this study. In designing the TCB, the electrode length of one subpixel of the TCB was designed to be 30% longer than that of the micro-LED. The gap between the p and n electrodes of the TCB was designed to be 30% longer than that between the p and n electrodes of the micro-LED, as shown in **Figure S1c** (Supporting Information). Therefore, in this case, the positional errors of the micro-LED

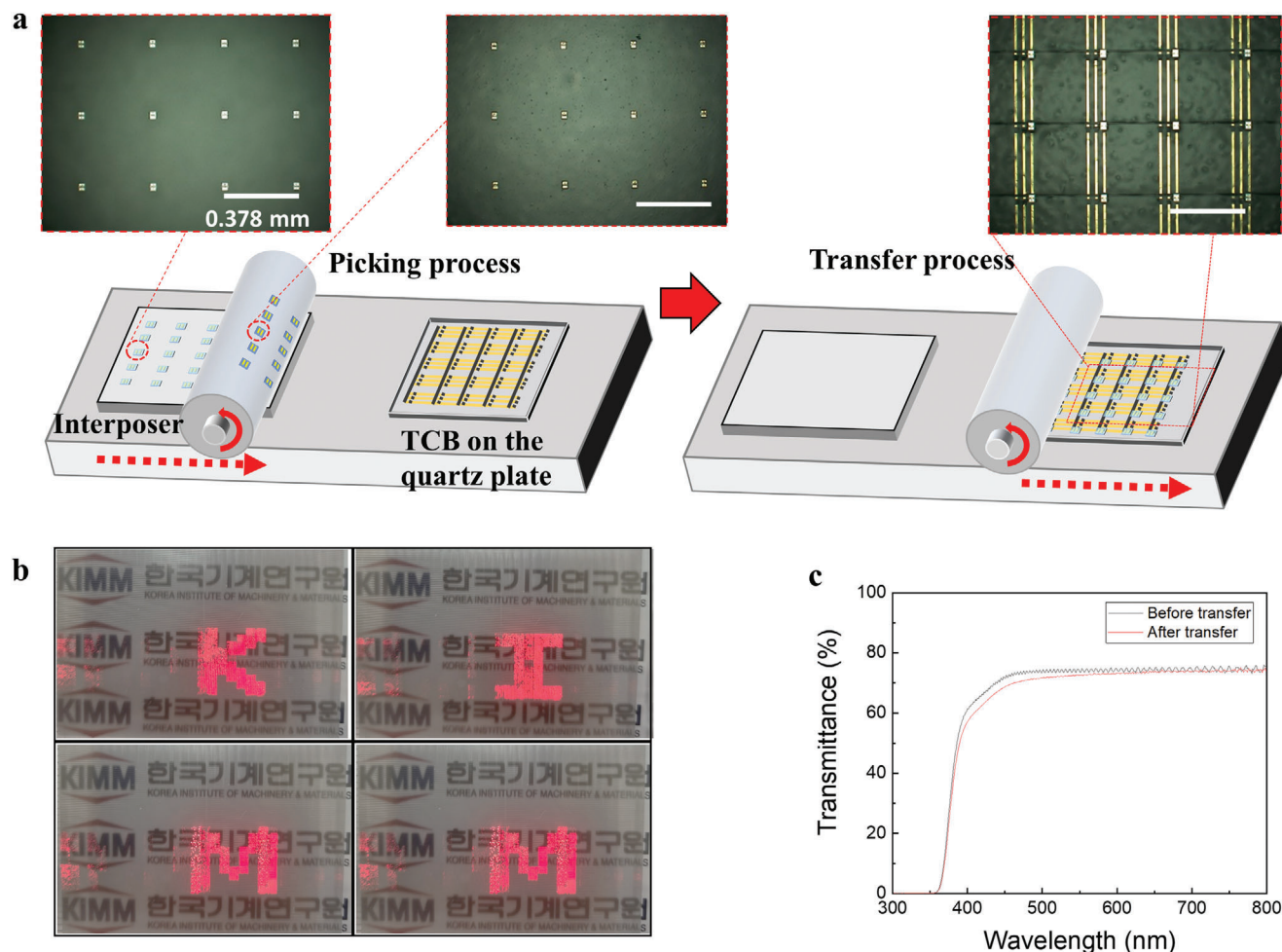


Figure 1. Roll-to-plate transfer process for transparent micro-LED displays. a) Schematic of transferring micro-LEDs on an interposer to a transparent circuit board (TCB) using a roll stamp. b) Demonstration of a transparent micro-LED display fabricated through the R2P transfer process. c) Transparency of the fabricated display, exhibiting 72.5% transmittance across the visible spectrum (450–700 nm).

were acceptable, with tolerances of 11 μm for the MD and 8.5 μm for the TD.

2.2. Realization of Roll-Based Overlay Alignment

The R2P transfer equipment developed in this study comprises a plate, quartz plate, upper camera, lower camera, and roll, as shown in Figure 2a. The plate is capable of movement in the x, y, and θ directions, and the upper and lower cameras can move in the x, y, and z directions. The roll can translate in the z-direction and rotate. The left and right sides of the roll move independently in the z direction, and there are load cells on both sides that receive load signals, enabling control of the load on each side.^[45] Furthermore, the translational velocity of the stage can be synchronized with the rotational velocity of the roll.

There are two methods for aligning the micro-LED array attached to the roll stamp and electrodes of the transparent substrate placed on the plate. The schematic of the calibrated alignment (CA) is presented in Figure 2b. The upper camera measured the electrode positions of the TCB, whereas the lower cam-

era measured the electrode positions of the micro-LED attached to the roll stamp, and the relative positions were then calculated. Subsequently, the x, y, and θ offset values of the plate are automatically input to align the positions, considering the error between the positions of the LED electrodes attached to the roll and the transparent PCB electrodes. This alignment method can lead to variations in the transfer repeatability, which is influenced by the repetitive precision of the rolling equipment hardware or software. Therefore, in this study, a RDOA method is proposed, as shown in Figure 2c. This approach is applicable when the PCB is transparent and uses a lower camera to align the electrodes of the micro-LED attached to the roll stamp and the TCB on the quartz plate.

Ten repeated tests were conducted to compare the iterative precision of CA and RDOA in the R2P equipment. As shown in Figure S6a (Supporting Information), the CA resulted in an MD iterative precision of 2.22 μm and a TD iterative precision of 2.59 μm . In the case of RDOA, as shown in Figure S6b (Supporting Information), the MD iterative precision is 0.52 μm , and that of the TD is 0.40 μm . Therefore, when transferring micro-LEDs to a TCB, RDOA enables the alignment

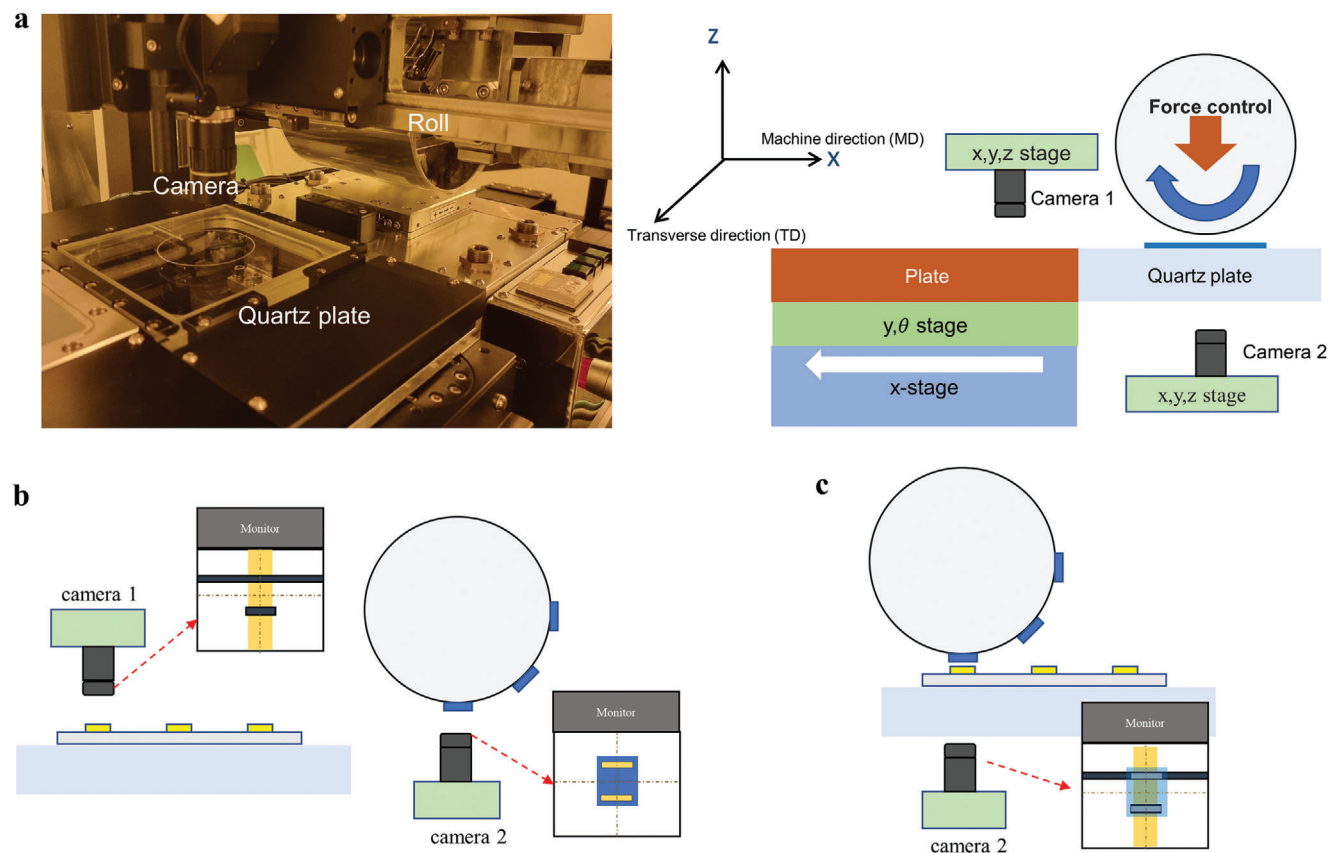


Figure 2. R2P equipment and alignment strategies. a) Image and schematic of R2P transfer equipment highlighting key components. b) Concept of calibrated alignment (CA) achieved by Camera 1 and Camera 2 for accurately positioning micro-LEDs. c) Concept of roll-based direct overlay alignment (RDOA) utilizing Camera 2 for adjustments to positioning in real-time.

of micro-LEDs and TCB electrodes with high precision within 1 μm .

2.3. Classification of Positional Errors in R2P Transfer Process

R2P transfer is a technology that is widely used in industry because of its high productivity; however, it has a disadvantage in that it is difficult to align the device with the substrate when transferring devices such as micro-LEDs. Therefore, for a successful R2P transfer process, it is important to classify the types of positional errors that occur in the process, and to figure out and resolve the reasons for each of them; three types of positional errors are defined in this study.

The first type of positional error is caused by misalignment of the micro-LED and TCB before the micro-LED is transferred to the TCB,^[1,3,38] as shown in Figure 3a. This commonly occurs in the CA process, resulting from an incorrect calculation of the relative positions of the micro-LED and TCB electrodes or when the repetitive precision of the R2P equipment is not optimal. Figure 3b shows a case of misalignment between the TCB and micro-LED electrodes. In the micro-LED transfer process, when there is an error in the x, y position at the starting point and a misalignment in θ with the ending point, the micro-LED is

transferred as shown in Figure 3b. As shown in Figure 3c, the second type of positioning error is the positional error of the TD, where the micro-LED is vertically misaligned on the TCB during the transfer process, even though the micro-LED and TCB are well-aligned before transferring the micro-LED. Figure 3d shows the occurrence of the TD error during micro-LED transfer.^[44] While the alignment is accurate at the beginning of the transfer, at the end of the transfer, the micro-LEDs deviate in the vertical direction of the TCB. The third type of positional error, shown in Figure 3e, is the positional error of the MD, where the pitch of the micro-LED array and that of the TCB are the same, but the position of the transferred micro-LEDs is misaligned in the horizontal direction of the TCB.^[39–43] Figure 3f shows an image of the micro-LED transfer when an MD error occurs. At the beginning of the TCB, there was no x-direction error for the micro-LEDs, but toward the end, an x-direction error occurred.

These types of positional errors are common in R2P transfers and are particularly noticeable in device transfers. Therefore, it is important to evaluate the transferred micro-LEDs and determine the exact cause of the positioning errors. In the following sections, experiments to determine the exact reason for each position error are introduced and solutions are presented.

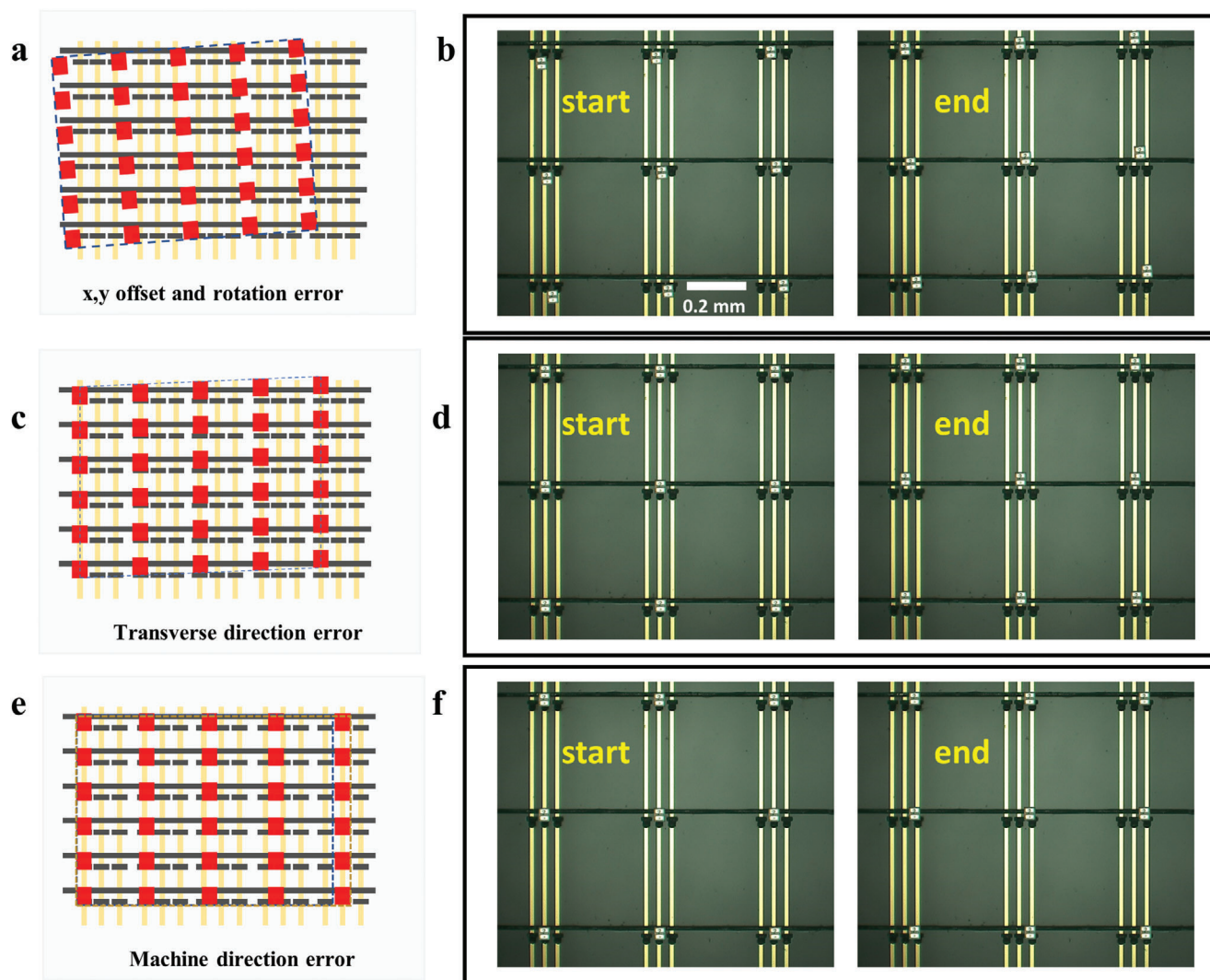


Figure 3. Classification of positional errors in R2P transfer process. a) Schematic of x,y offsets and rotation errors during transfer process. b) Image of x,y offset and rotation errors at the start and end points of the TCB. c) Schematic of transverse direction (TD) error, where misalignment occurs despite initial device-PCB alignment. d) Image of TD error at the start and end points of the TCB. e) Schematic of error at the start and end points of the TCB. f) Image of MD error at the start and end points of the TCB.

2.4. Quantitative Evaluation of TD Positional Error

Despite achieving precise alignment of micro-LEDs and TCB through the RDOA process, TD errors occur during the transfer process, causing misalignment of the LEDs in the vertical direction of the rolling progression. The main cause of the TD errors is the change in the displacement difference between the left and right sides of the roll during the micro-LED transfer process. To validate this, tests were performed for three different cases. In this test, the left (z1)–right (z2) displacement difference of the roll was zero, indicating that the left-right tilt of the roll and the left-right tilt of the TCB were the same. Load control is the ability to maintain a constant load and transfer it based on signals from the left and right load cells at the initial contact.

Figure 4a shows a schematic of the experiment when there is a non-zero z1–z2 displacement difference in the roll during the initial contact, with the implementation of load control. The graph

in Figure 4b shows that the load starts at 3.9 kgf in z1 and 3 kgf in z2 and is controlled to 3.5 kgf by the load control, and the displacement difference of the roll changes from 27 to 11 μm from the start. Figure 4c presents images captured by the bottom camera of the roll and shows that the position of the pattern is also shifted toward z2 at the end point.

Figure 4d shows a schematic of the experiment when there is a zero z1–z2 displacement difference in the roll during the initial contact, with load control. The z1–z2 displacement of the roll was controlled to match the 3.5 kgf load on both sides of the roll, with a slight change in the displacement difference, as shown in Figure 4e. Figure 4f shows that the TD position of the pattern at the starting point and at the endpoint did not change.

Figure 4g shows a schematic of the experiment when there is a nonzero z1–z2 displacement difference in the roll during the initial contact, without load control. As there was no load control, the load signals from the load cells on both sides of the roll were

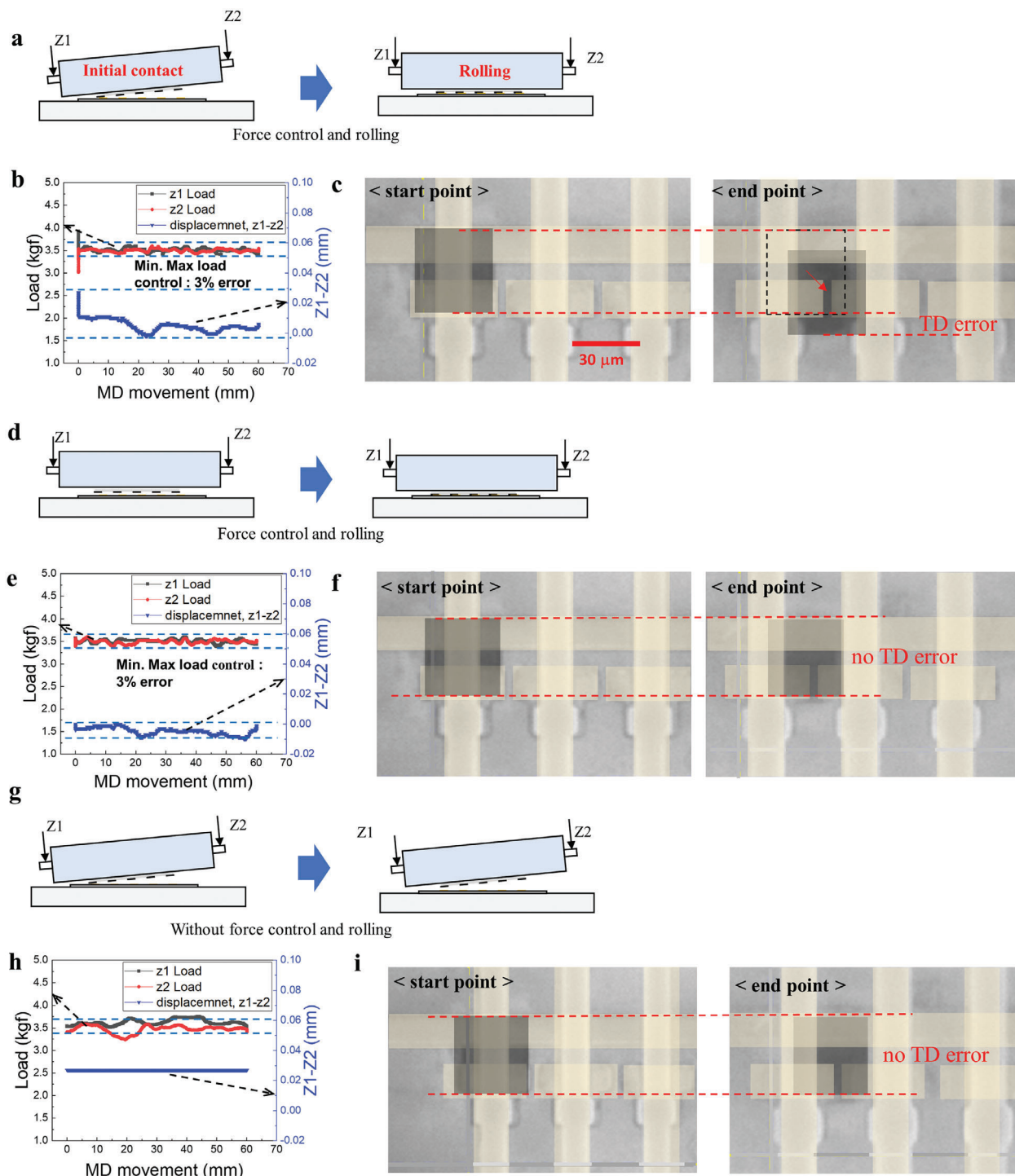


Figure 4. Quantitative evaluation of TD positional error. a) Schematic of the experiment for investigating the impact of miss-initial contact and adjusting load control on TD Error. b) Graph of load variation of Z1, Z2 and displacement difference of Z1, Z2 in case of miss-initial contact and adjusting load control. At the starting point, the Z1 and Z2 loads are controlled from 3.9 kgf and 3.0 kgf to 3.5 kgf, respectively, and the displacement difference of Z1 and Z2 is changed from 27 to 11 mm. c) Image of the positional error of TD in case of miss-initial contact and adjusting load control. d) Schematic of the experiment for investigating the impact of good-initial contact and adjusting load control on TD Error. e) Graph of load variation of Z1, Z2 and displacement difference of Z1, Z2 in case of good-initial contact and adjusting load control. The load on Z1, Z2 is controlled at 3.5 kgf starting from 3.5 kgf, and the displacement difference of Z1, Z2 is small. f) Image of the positional error of TD in the case of good-initial contact and adjusting load control. No positional errors of TD. g) Schematic of the experiment for investigating the impact of miss-initial contact and no-load control on TD Error. h) Graph of load variation of Z1, Z2 and displacement difference of Z1, Z2 in case of miss-initial contact and no-load control. The load on Z1, Z2 is uncontrolled, resulting in large variations, but the displacement of Z1, Z2 is constant. i) Image of the positional error of TD in case of miss-initial contact and no-load control. No positional errors of TD.

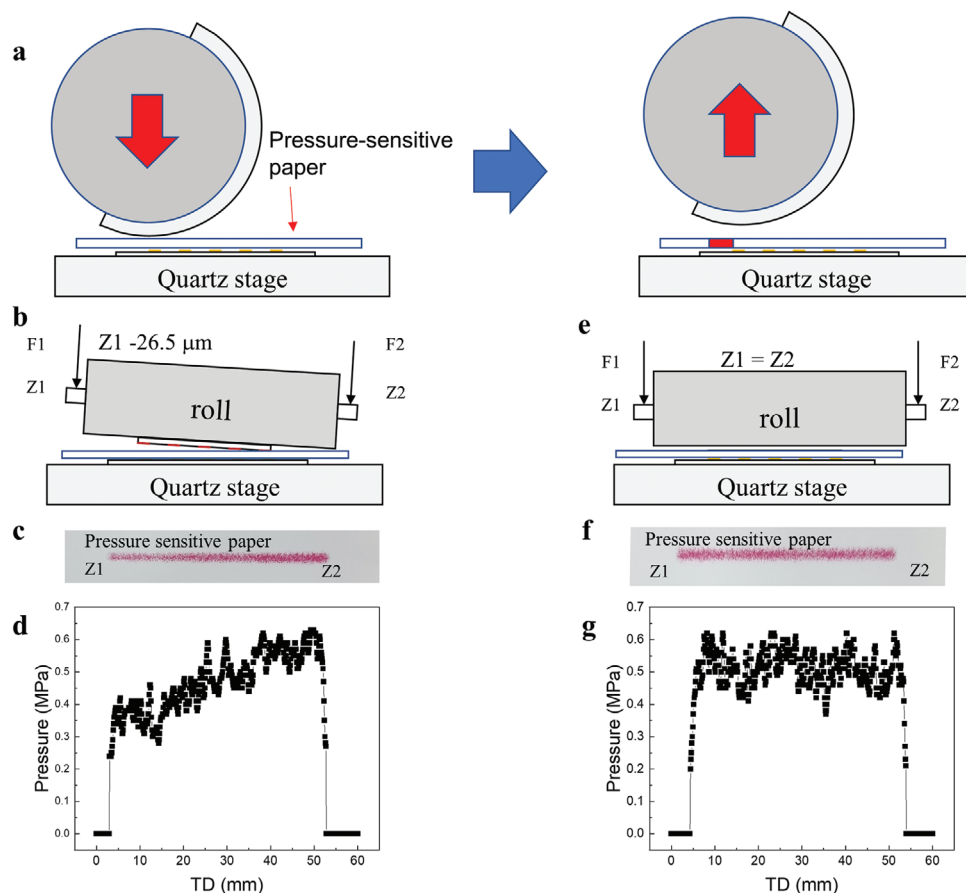


Figure 5. Method for making initial contact in R2P transfer process. a) Schematic of method for initial contact experiment between roll stamp and TCB using pressure-sensitive paper. b) Initial contact schematic with roll stamp's Z1 displacement $26.5 \mu\text{m}$ relative to Z2 displacement and reaction forces F1 and F2 due to contact between roll and plate. c) When the Z1 displacement of the roller stamp above the Z2 displacement by $26.5 \mu\text{m}$, the pressure distribution formed on the pressure-sensitive paper shown in (c), non-uniformity in pressure distribution. d) The measured pressure distribution from the pressure-sensitive paper shown in (c), non-uniformity in pressure distribution. e) Initial contact schematic with Z1 and Z2 displacements being equal. f) When the Z1 displacement of the roll stamp and the Z2 displacement are equal, the pressure distribution formed on the pressure-sensitive paper. g) The measured pressure distribution from the pressure-sensitive paper shown in (f), uniformity in pressure distribution.

not regulated, whereas the displacement difference on both sides of the roll was constant, as shown Figure 4h. The TD position of the pattern at the starting point and that at the endpoint did not change, as shown in Figure 4i. Because the positional errors of the TD can be confused with misalignment errors, it is important to determine the exact type of error that is generated. Therefore, it is essential to understand the appearance and identify the reason for the TD position errors. From the results of these experiments, it can be concluded that the positional error of the TD was caused by the difference between the z1 and z2 displacements of the roll at the initial contact and the z1 and z2 displacement of the roll during the rolling process. This problem can be solved by eliminating the source of the positional error. When the relative tilt between the roll stamp and TCB is precisely matched before the micro-LED transfer, the corresponding pressure between the roll and TCB can be kept constant by load control during the transfer process. This uniform contact pressure eliminates the TD error.

To determine the initial contact displacement of the roll stamp, we utilized pressure-sensitive paper (LLW, Prescale, Fuji film),

as shown in Figure 5. Placing pressure-sensitive paper between the roll stamp and TCB and applying load results in pressure distribution on the paper, as shown in Figure 5a. Figure 5b represents the experiment conducted after raising the Z1 side of the roll stamp by $26.5 \mu\text{m}$. F1 and F2 are measured by the load cells attached on the Z1 and Z2 sides respectively. After the roll and plate are in contact and the roll is moved up, the pressure distribution is measured on the pressure-sensitive paper as shown in Figure 5c. When converted into pressure for analysis, the pressure is non-uniform, with the pressure on the Z1 side being lower as shown in Figure 5d. Figure 5e provides a schematic of the matched tilt between the roll stamp and plate, while Figure 5f illustrates the pressure distribution on the pressure-sensitive paper. The measured pressure distribution is uniform as shown in Figure 5g indicating successful initial contact. Through this experiment, we were able to determine the initial contact displacement of the roll and eliminate the cause of the TD positional errors. Also, this solution can easily be applied to the R2P transfer process to remove position errors in the TD.

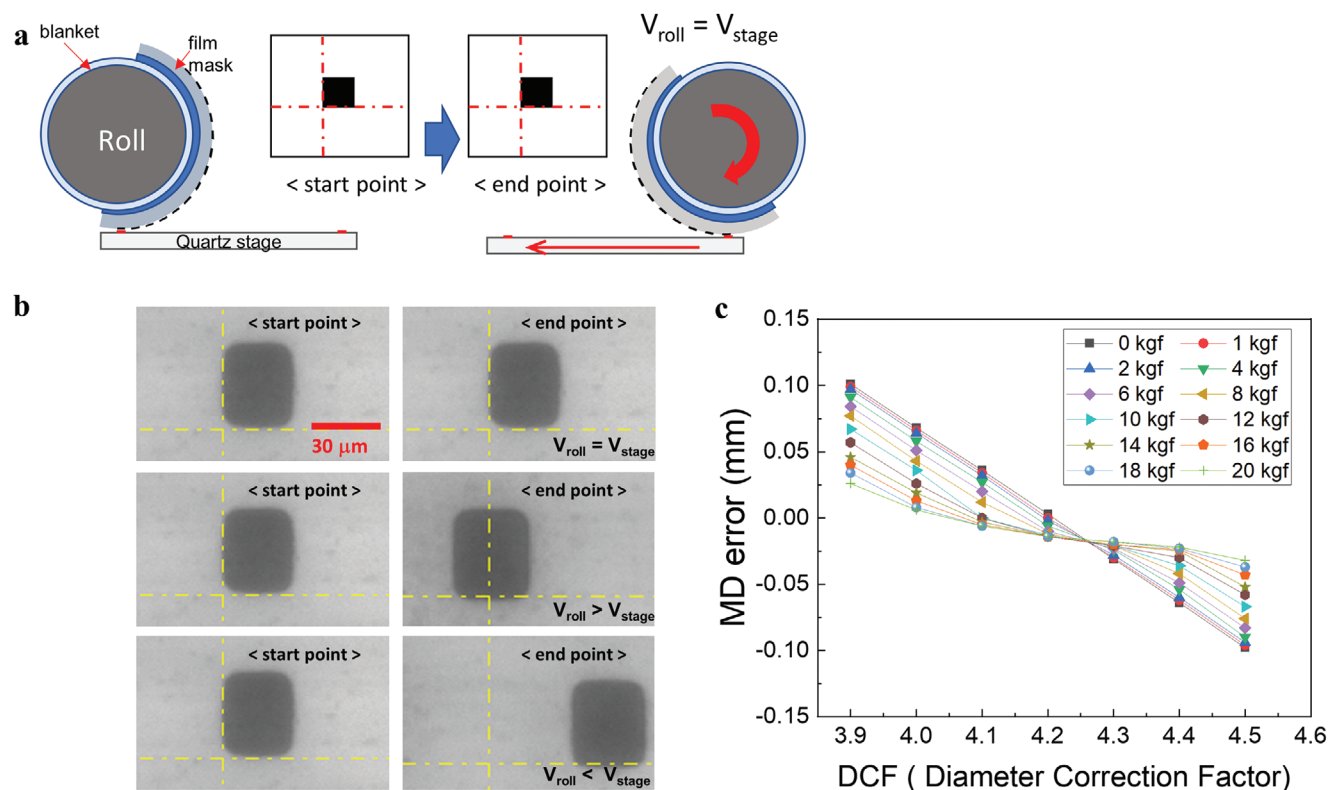


Figure 6. Quantitative evaluation of MD positional error. a) Schematic of MD error at identical roll and stage velocities. b) Experiment images of MD error under varying roll and stage velocities. c) Relationship between MD error and diameter correction factor (DCF) based on load measurements.

2.5. Quantitative Evaluation of MD Positional Error

The MD error is a phenomenon in which the pitch of the micro-LED array attached to the roll stamp changes, despite being equal to the electrode pitch of the TCB. This discrepancy occurs because of the asynchronous linear velocities of the roll and stage. To analyze the MD error, a film mask was used instead of actual micro-LEDs. The pitch of the film mask was designed considering the bending strain when the film was attached to a roll, as shown in Figure S7a (Supporting Information). As shown in Figure S7b (Supporting Information), the pattern size was $32 \mu m \times 40 \mu m$, with a pitch of 0.126 mm. The roll stamp had a radius of 75 mm and was wrapped with a blanket, under a blanket, and in film, each with thicknesses of 0.3, 0.07, and 0.188 mm, respectively. When the film was wrapped around the roll, a bending strain of 0.12% occurred. Therefore, the pitch of the fabricated patterned film was 0.12% smaller than that of the TCB film. Figure 6a shows the experimental plan for synchronizing the rotational velocity of the roll with the translational velocity of the plate. When the rotational velocity of the roll was equal to the translational velocity of the plate, a pattern appeared at the start and end points, as shown in Figure 6a. Figure 6b shows the images depicting the position of the pattern when the roll's rotational velocity was the same as the stage's velocity, when the roll velocity was faster, and when the roll velocity was slower. When the roll and stage velocities were identical, the pattern stopped at the exact position. However, when the roll velocity was higher, the pattern stopped quicker at the endpoint, thus reducing the

pattern's pitch. Conversely, if the roll velocity was lower, the pattern appeared to stop later than the stage, thereby increasing the pattern's pitch. Figure 6c presents the MD error values based on the roll diameter correction factor (DCF) for different loads. With the same roll DCF, as the load increased, the rotational velocity of the roll decreased, and the pattern's pitch increased.

The positional error value of the MD can vary as a load is applied, even if the diameter of the roll stamp is fixed. As the material of the roll stamp is deformed by the load, the diameter of the roll stamp changes, affecting its rotational velocity. This means that when more force is applied to the roll stamp, the diameter of the roll stamp decreases and the rotation velocity of the roll decreases. Therefore, it is necessary to correct the roll stamp diameter, which is reduced by the load. To solve this problem, in this study, the DCF was used to correct the roll stamp diameter. The positional error in the MD can then be solved by determining the DCF per load.

2.6. R2P Process Flow for High-Precision and High-Yield Transfer of Micro-LEDs

In this section, we present a process flow designed to tackle the inherent alignment challenges associated with the roll-to-plate (R2P) transfer of micro-LEDs. The flowchart in Figure S8 (Supporting Information) provides a concise overview, and the key steps are outlined as follows: To achieve high-yield micro-LED transfer, we first determine the appropriate transfer load based

on the number of micro-LEDs and the target transfer area. Once the transfer load is established, we calculate the corresponding “Roll Diameter Correction Factor” using data from Figure 6c. By assigning an initial estimate for the roll displacement values (Z_1 and Z_2), the roll is moved into contact with the TCB. The resulting pressure distribution is measured using a pressure-sensitive paper as described in Figure 5. When the pressure distribution is not uniform, we update the roll displacement values (Z_1 and Z_2) accordingly. When the pressure distribution becomes uniform, the preparation for the R2P transfer process is complete. The transfer load and the associated “Roll Diameter Factor” are assigned as control parameters for the transfer machine. This step helps eliminate MD errors. Using the proposed RDOA method, we secure precise positional alignment between micro-LEDs on the roll and the TCB on the transparent stage. The roll moves toward the TCB until the prescribed transfer load is reached. If the load of Z_1 and Z_2 is different, we adjust the roll displacement until the load of Z_1 and Z_2 is the same. Once this criterion is met, the transfer process proceeds. By following this process flow, we anticipate achieving both high yield and high precision in micro-LED transfer using the R2P method.

3. Conclusion

We have studied the roll-to-plate (R2P) transfer technology for mass production of micro-LEDs. R2P transfer is a promising technique for achieving high-yield transfer, large-area transfer, continuous processing, and minimal micro-LED damage. However, despite these merits, the nature of roll-based transfer poses challenges in accurately placing micro-LEDs onto the electrodes of a transparent circuit board (TCB). In this study, we proposed a Roll-based Direct Overlay Alignment (RDOA) method to address this issue. This method, applicable when the final substrate and plate are transparent, allows for alignment within 1 μm of repeatability by simultaneously measuring the positions of micro-LED electrodes and TCB electrodes for alignment during the transfer process. Despite precise X-Y-theta alignment, gradual positional errors of micro-LEDs can still occur and accumulate during the roll transfer process. We differentiate these gradual errors into Machine Direction (MD) and Transverse Direction (TD) positional errors, and provide respective causes and solutions. MD positional errors result from asynchronous linear velocities between the roll and stage. Through experimental validation, we obtained the “Roll Diameter Correction Factor” for different loads, effectively eliminating MD errors by applying this factor based on the transfer load. TD errors arise from variations in the Z_1 - Z_2 displacement of the roll during initial contact and the roll transfer process, and we can mitigate the TD errors by ensuring a uniform contact pressure distribution during both initial contact and the entire roll transfer process. To demonstrate practical feasibility, we proposed a process flow for a highly accurate RDOA method and successfully transferred 30 μm -sized micro-LEDs in a 160×120 array using our R2P equipment. The proposed roll-based transfer method achieved a transfer yield of over 99.75% and a maximum positional error of 3.2 μm . The reason for the maximum error of 3.2 μm is that the positional error of the micro LEDs arranged on the interposer is $\approx 3 \mu\text{m}$. Our study addresses alignment challenges in the R2P process and contributes to the commercialization of transparent micro-LED displays. It is

expected to positively impact on the manufacturing of transparent applications that involve rolling processes.

4. Experimental Section

R2P Transfer Process of Micro-LEDs and Reflow Process: In this study, the R2P transfer process was used to transfer micro-LEDs grown on sapphire substrates for use as pixels in transparent displays. Micro-LEDs were arranged as interposers on a sapphire substrate using a laser lift-off process. The interposer was polydimethylsiloxane (PDMS) coated on a 4-inch silicon wafer. The interposer of the arranged micro-LEDs was placed on the hot plate of the R2P equipment to pick it up on the roll stamp. The temperature of the hot plate was set to 50 $^{\circ}\text{C}$ and the picking load was set to 18 kgf. The temperature was maintained during picking to achieve uniform contact between the roll stamp and interposer. After picking up the micro-LED with a roll stamp, the flux-coated TCB was placed on a transparent quartz plate for temporary bonding between the micro-LED and TCB. Because the quartz plate and TCB are transparent, the electrodes of the TCB and micro-LED on the roll stamp can be aligned using the lower camera of the R2P. After aligning the micro-LED on the roll stamp with the electrodes on the TCB, the roll was moved down in the z-direction, and a load of 18 kgf was applied. The micro-LEDs were transferred to the TCB as a load was applied and rolled. After the transfer of the micro-LEDs, the reflow process was conducted for the electrical connection of the micro-LED electrode and TCB electrode, using commercial equipment (x-reflow306), and the process was conducted at a peak temperature of 280 $^{\circ}\text{C}$.

Finite Element Analysis (FEA) of Nip Width of R2P: To transfer micro-LEDs using roll-to-plate equipment, it was essential to measure the transfer pressure accurately. In the R2P process, when the roll and plate were in contact, the deformation of the blanket wrapped around the roll created a contact area. The contact area could be obtained by multiplying the nip width by the width of the transfer film at the contact between the roll and plate. The tip width varied with the load, and as the load increased, the nip width increased nonlinearly. Therefore, knowledge of the nip width for each load was crucial to accurately measure the transfer pressure. In this study, Abacus software was used to model the contact between the roll stamp and quality plate, and the results were compared with the experimental results. The roll stamp was modeled as a quarter to reduce the analysis time and was composed of a stainless-steel roll, PDMS blanket, PET under blanket, PET transfer film, and quartz plate material, as shown in Figure S9a (Supporting Information). The properties of the materials used in the FEA are listed in Table S2 (Supporting Information). The roll, blanket, under blanket, and transfer film are tightly bonded, and the roll stamp and quartz plate are in contact. Using general contact. The load at the contact surface and the nip width were measured by applying a displacement to the roll stamp. The analysis results show nip widths of 1.31, 2.75, and 3.39 mm at loads of 2.39, 9.99, and 15.6 kgf, respectively, as shown in Figure S9b (Supporting Information). As shown in Movie S2 (Supporting Information), deformation occurred in the PDMS blanket, with tensile deformation in the contact areas and compressive deformation in the non-contact areas. However, the strain of the transfer film, where the roll and plate are in direct contact, is constant, indicating that the positional error of the MD is linear when transferring micro-LEDs. In addition, when two objects are in contact, there is no uniform stress distribution at the contact surface. However, in the case of the roll stamp used in this analysis, the blanket causes the stress to be uniform at the contact, and the area of uniform stress increases as the load is applied. The FEA analysis of the roll stamp and quartz plate used in the actual experiment showed trends in the nip width, stress, and strain distribution. Even if the properties or materials of the roll stamp change, it is possible to predict characteristics such as the nip width, stress, and strain distribution.

Nip Width Measurement of R2P: FEA can predict the nip width owing to the contact of the roll stamp with the stage under a load. However, it is necessary to verify that the modeling in FEA is correct, and an experiment was conducted to evaluate the contact between the roll stamp and quartz

stage. The nip width of the roll stamp of the R2P equipment was measured as shown in Figure S9c (Supporting Information). The experimental results indicated that the nip width was 1.37 mm at 2 kgf, 2.90 mm at 10 kgf, and 3.40 mm at 16 kgf and above. As shown in Figure S9b (Supporting Information), the FEA and experimental data are not perfectly equal but show similar trends.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

micro-LED, overlay alignment, roll to plate, transfer, transparent display

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